

Progress Update: Revisions After Professor Feedback

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Outline

- 1 Professor's Suggestions
- 2 Metric Orthogonality
- 3 Separating Feature Selection from Prediction
- 4 Separating Regression from Classification





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Key Suggestions from Meeting

■ Metric orthogonality

- Identify collinear metrics via pairwise correlation; keep a minimal orthogonal set
- Move redundant metrics to appendix

■ Separate Feature Selection from Prediction

- Mixing different FS + different classifiers = comparing apples with pears
- Run two controlled experiments: fix one, vary the other

■ Separate Regression from Classification

- Two distinct problem formulations; must not be mixed in the same table

■ Show selected features in figures

- Each FS method result should illustrate which features were actually chosen



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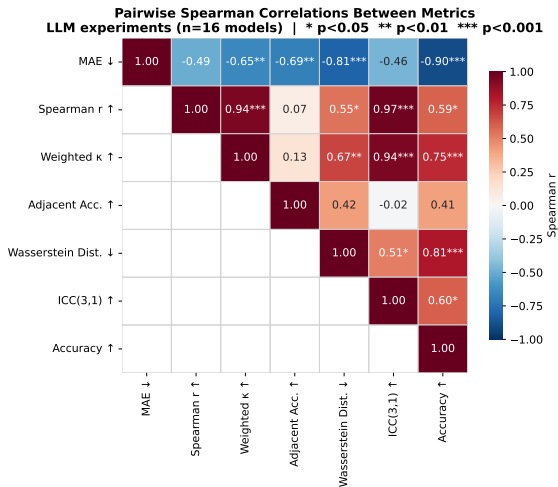
Metric Orthogonality – Approach

- Problem: too many metrics confuse readers; many carry the same information
- Approach: pairwise Spearman correlation + hierarchical clustering + PCA
- Data: LLM experiments ($n=12$ models, 6 metrics) and ML experiments ($n=104$, 4 metrics)

Candidate metrics: MAE, Spearman ρ , Weighted κ , Adjacent Acc., Kendall τ_b , ICC(3,1)

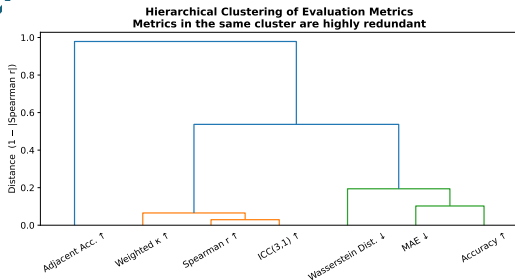


Pairwise Spearman Correlations (LLM data, $n=12$)





Hierarchical Clustering of Metrics



3 clusters found

(threshold $|r_s| > 0.75$):

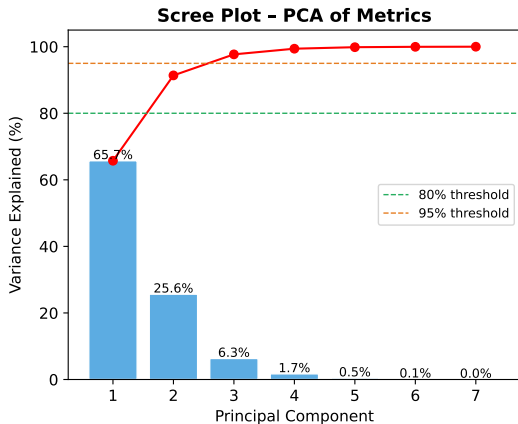
- **Cluster 1:** Spearman ρ , Weighted κ , Kendall τ_b , ICC(3,1)
- **Cluster 2:** MAE, Binary F1
- **Cluster 3:** Adjacent Acc. (*most independent*)

PCA: PC1 = **78.7%** variance
⇒ highly redundant



PCA of Metrics

Principal Component Analysis of Evaluation Metrics
LLM experiments (n=16) — metrics standardised, MAE inverted



PCA Loadings (first 4 PCs)

MAE ↓	0.38	0.43	0.13	0.19
Spearman r ↑	0.39	-0.34	-0.38	0.12
Weighted κ ↑	0.44	-0.23	-0.16	-0.01
Adjacent Acc. ↑	0.20	0.63	-0.38	0.42
Wasserstein Dist. ↓	-0.40	-0.30	-0.01	0.84
ICC(3,1) ↑	0.39	-0.39	-0.18	0.09
Accuracy ↑	0.39	-0.07	0.80	0.26
	PC1	PC2	PC3	PC4



Final Metric Set



Retained (4 orthogonal metrics):

- **MAE** $\downarrow$ — absolute error magnitude
- **Spearman** $\rho \uparrow$ — rank correlation; represents Cluster 1
- **Weighted** $\kappa \uparrow$ — ordinal agreement; clinical standard
- **Adjacent Acc.** $\uparrow$ — clinically meaningful tolerance; most independent

Dropped (moved to appendix):

- Kendall τ_b $r_s=0.993$ with Spearman
- ICC(3,1) $r_s=0.958$ with Kappa
- Binary F1 correlated with MAE

Pair	r_s
Spearman vs Kendall	+0.993***
Kappa vs ICC	+0.958***
Spearman vs Kappa	+0.839***
Spearman vs AAcc	+0.274



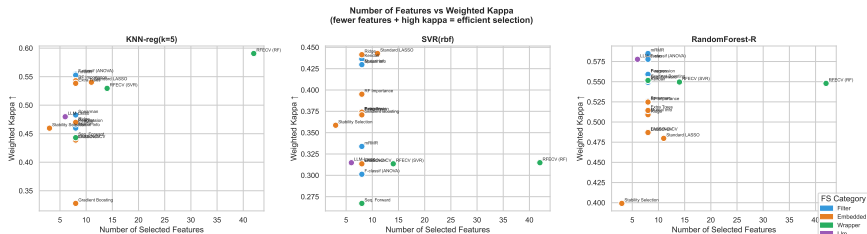
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Experiment A – Feature Count vs. Kappa





Experiment B – Compare Predictors (Fixed FS)

- Fixed FS method: **LLM-Lasso** (LLM-based, 6 features selected)
- 26 predictors compared: 13 regressors + 13 classifiers (separated)
- Best regressor: **GradBoost-R** (Pareto-dominant across all metrics)
- Best classifiers: **KNN-clf** and **ExtraTrees-C** (Pareto-optimal, each excels on different metrics)



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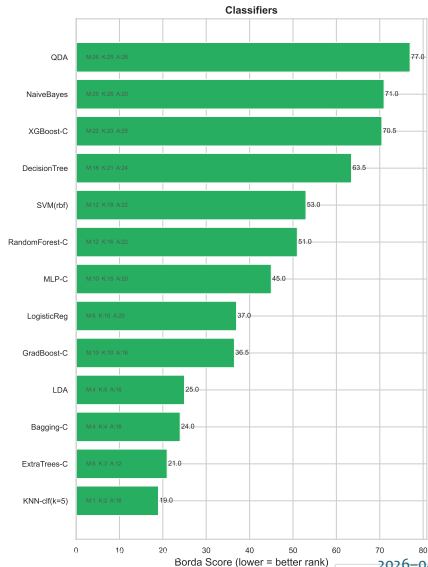
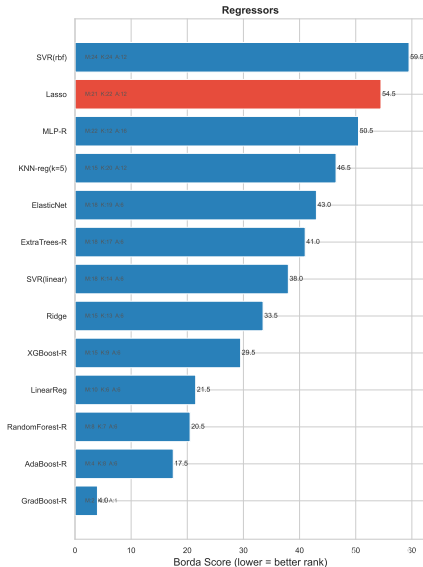
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Borda Count Ranking – All Predictors (LLM-Lasso, 6)

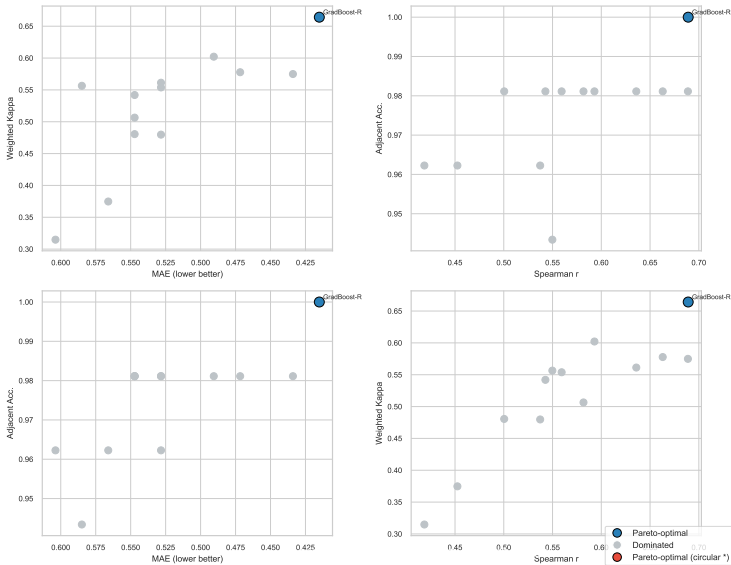
Borda Count Ranking – LLM-Lasso Features (6 features, LOOCV n=53)
Per-metric ranks summed: MAE(M) Kappa(K) AAcc(A) | lower = better





Pareto Dominance – Regressors

Pareto Dominance Analysis – Regressors (LLM-Lasso, 6 features, LOOCV n=53)
Pareto-optimal: GradBoost-R

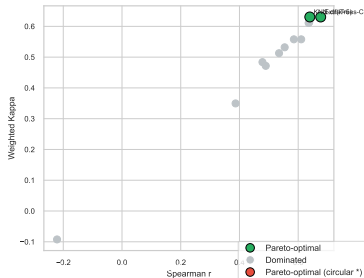
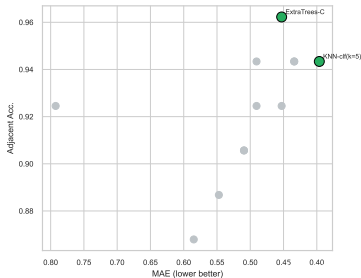
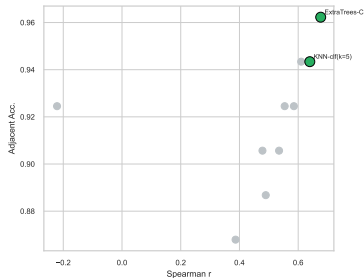
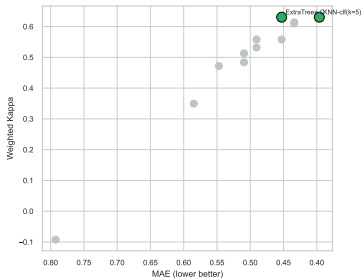




Pareto Dominance – Classifiers

Pareto Dominance Analysis – Classifiers (LLM-Lasso, 6 features, LOOCV n=53)

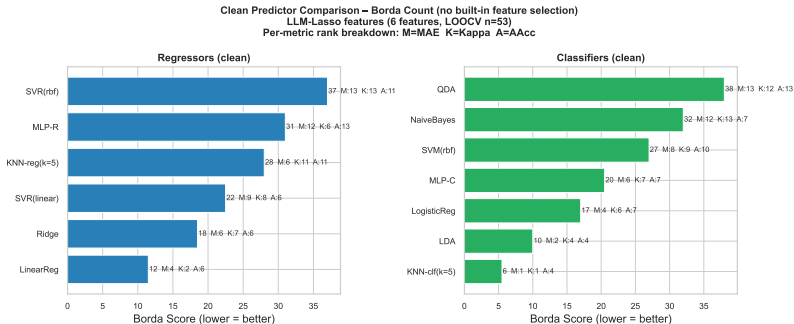
Pareto-optimal: KNN-clf(k=5), ExtraTrees-C





Clean Predictor Comparison (No Built-in Feature Selection)

- Excludes models with internal feature weighting (RF, GradBoost, XGBoost, LASSO...)





Circular Combinations – Flagged with *

- **Definition:** FS method uses the same algorithm as the predictor
 - RF Importance FS + RandomForest-R/C
 - RFECV-RF FS + RandomForest-R/C
 - LLM-Lasso FS + Lasso regressor
- **Problem:** cannot distinguish whether performance is from FS or model
- **Treatment:** included but marked with * and red border in figures

